

## Where we left off...

- Every SISO-LTI-system can be written as a fraction of two polynomials in frequency domain.

### Time domain

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

Laplace-Transform

$$G(s) = C(sI - A)^{-1}B + D$$

### Frequency domain

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0} + d$$

- The variable  $s$  is a "complex frequency".  
 $s = \sigma + j\omega$ , where

- $\text{Re}(s) = \sigma$ : exponential part

- $\text{Im}(s) = \omega$ : sinusoidal part

$\Rightarrow$  In this course, most of the time we have  
 $\sigma = 0$  (pure sinusoids).

• We saw, that most signals can be written as

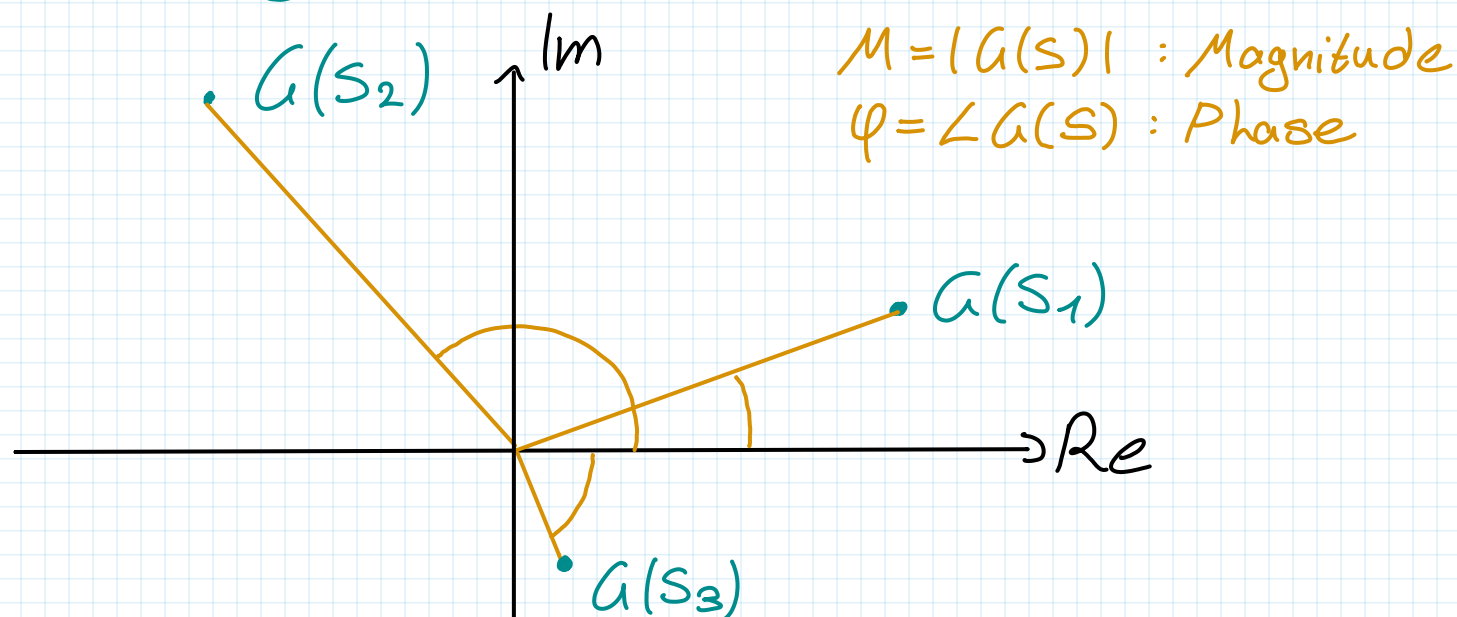
$$u(t) = \sum_i \underline{U_i} e^{\underline{s_i} t} \Rightarrow \text{Ex: } \cos(\omega t) = \underline{\frac{1}{2}} (\underline{e^{j\omega t}} + \underline{e^{-j\omega t}})$$

• Since we had that  $u \rightarrow \boxed{G(s)} \rightarrow y$ , we find that

$y_{ss}(t) = \sum_i G(s_i) U_i e^{s_i t}$ , but if we evaluate this for some  $s = \sigma + j\omega$ , we get weird complex values...

$$\Rightarrow y_{ss}(t) = \underline{G(j\omega)} e^{j\omega t} + \underline{G(-j\omega)} e^{-j\omega t}$$

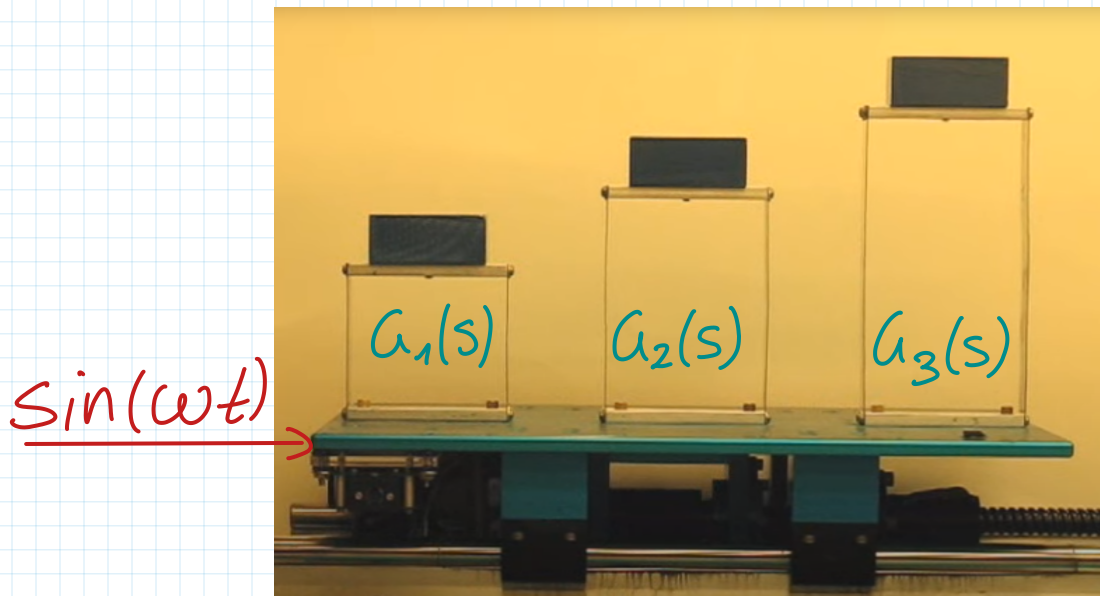
$\Rightarrow$  Plotting this for different  $s$  we can find:



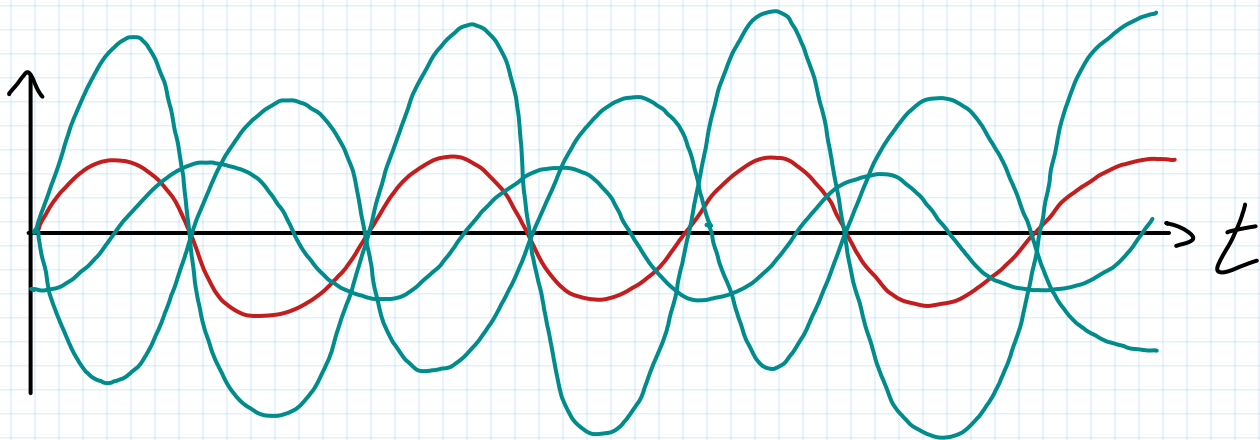
$$\begin{aligned} \Rightarrow \text{We can write } y_{ss}(t) &= \underline{\frac{1}{2} M e^{j\varphi}} e^{j\omega t} + \underline{\frac{1}{2} M e^{-j\varphi}} e^{-j\omega t} \\ &= M \cdot \underline{\frac{1}{2}} (e^{j(\omega t + \varphi)} + e^{-j(\omega t + \varphi)}) \end{aligned}$$

$\Rightarrow$  which is just  $y_{ss}(t) = M \cos(\omega t + \varphi) !!!$

# Intuition



$\Rightarrow$  For the same  $s$ , these systems have different responses.



- Inputs experience a change in
  - Amplitude:  $|G(s_i)|$
  - Phase:  $\angle G(s_i)$ ,
 depending on input frequency.

$\Rightarrow$  Output frequency does not change!!

Now we have fully characterized systems in frequency domain right? No...

• Take another look at the formula for the transfer function  $G(s)$ ...

$$G(s) = \underline{C}(s\mathbf{I} - \underline{A})^{-1}\underline{B} + \underline{D}$$

$\Rightarrow$  Notice how we said a LTI-system is fully characterized by the following equation:

$$\underbrace{y_{IC}(t)}_{\text{homogeneous}} = \underline{C}e^{\underline{A}t}\underline{x}(0) + \underbrace{\int_0^t \underline{C}e^{\underline{A}(t-\tau)}\underline{B}u(\tau)d\tau + \underline{D}}_{\text{particular}} = y_F(t)$$

$\Rightarrow$  All parameters included in TF formula, except:

$x(0)$  !!

$\Rightarrow$  TF's only show the forced response!

So far so good?

Let's dive in-depth into TF-structure

## Transfer Functions: Part II

- Standard form:

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0} + d$$

- Root-Locus form:

$$G(s) = \frac{k_H \cdot (s - z_1)(s - z_2) \dots (s - z_m)}{s^q (s - p_1)(s - p_2) \dots (s - p_{n-q})}$$

- Bode form:

$$G(s) = \frac{k_{\text{Bode}}}{s^q} \cdot \frac{\left(\frac{s}{-z_1} + 1\right)\left(\frac{s}{-z_2} + 1\right) \dots \left(\frac{s}{-z_m} + 1\right)}{\left(\frac{s}{-p_1} + 1\right)\left(\frac{s}{-p_2} + 1\right) \dots \left(\frac{s}{-p_{n-q}} + 1\right)}$$

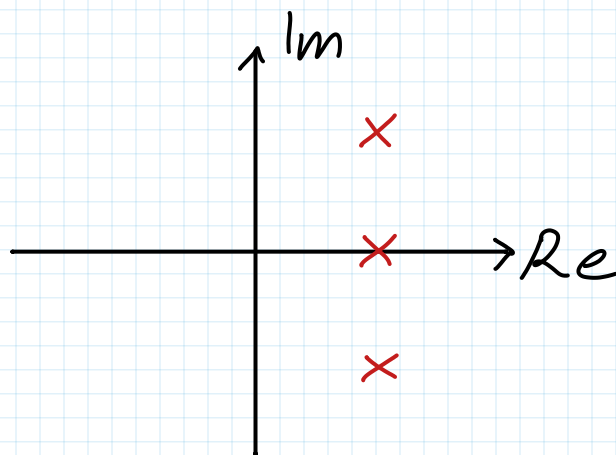
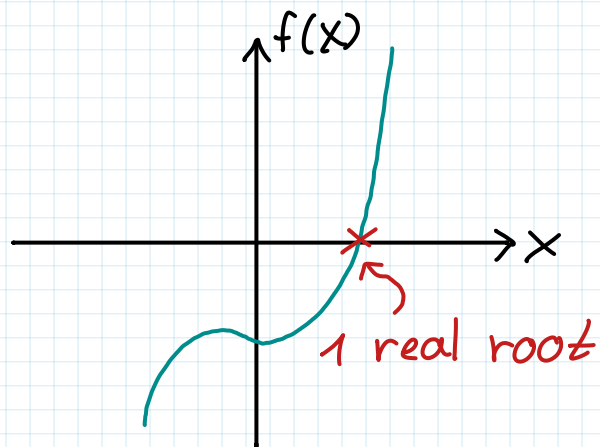
- The scalar values  $p_i$  are called **poles** of  $G(s)$
- The scalar values  $z_i$  are called **zeros** of  $G(s)$

Def: Poles  $\Rightarrow$  roots of the denominator of  $G(s)$   
 Zeros  $\Rightarrow$  roots of the numerator of  $G(s)$

$$G(s) = \frac{N(s)}{D(s)} \Rightarrow \begin{array}{l} z_i: s_i \text{ s.t. } N(s) = 0 \\ p_i: s_i \text{ s.t. } D(s) = 0 \end{array}$$

- Fundamental theorem of algebra:  
 Order of polynomial  $\hat{=}$  # (complex) roots

Ex:  $f(x) = x^3 - x^2 - 2 = 0 \Rightarrow (x-1)(x+1+j)(x+1-j) = 0$

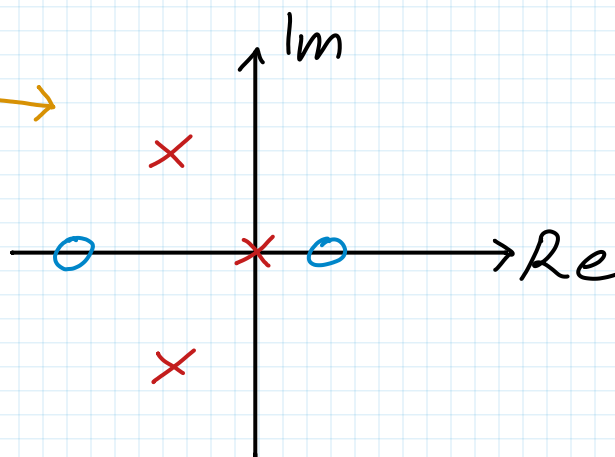


$\Rightarrow$  The location of poles & zeros are the defining characteristic of an LTI-system.

- Graphical notation (given root-locus form)

$$G(s) = \frac{k \cdot \textcolor{blue}{(s+2)} \cdot \textcolor{blue}{(s-1)}}{\textcolor{red}{s} \cdot \textcolor{red}{(s+1+j)} \cdot \textcolor{red}{(s+1-j)}}$$

$\textcolor{red}{x}$ : poles,  $\textcolor{blue}{o}$ : zeros



## Poles

Recall:  $G(s) = C(s\mathbb{I} - A)^{-1}B + D$

From linalg, we had that  $A^{-1} = \frac{\text{adj}(A)}{\det(A)}$

$$\Rightarrow G(s) = \frac{1}{\det(s\mathbb{I} - A)} \cdot \overbrace{C \cdot \text{adj}(s\mathbb{I} - A)B + D}^{\textcolor{red}{N(s)}} = \frac{N(s)}{\det(s\mathbb{I} - A)}$$

$\Rightarrow$  Setting  $\det(sI - A) = 0$  is how we find eigenvalues!

Poles of  $G(s) \hat{=}$  Eigenvalues of  $A$

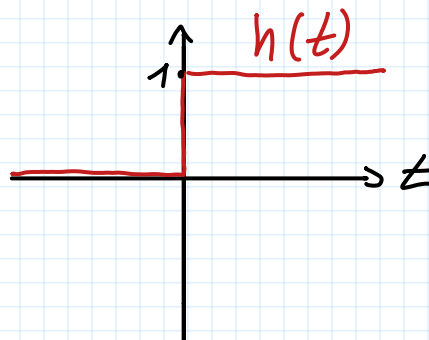
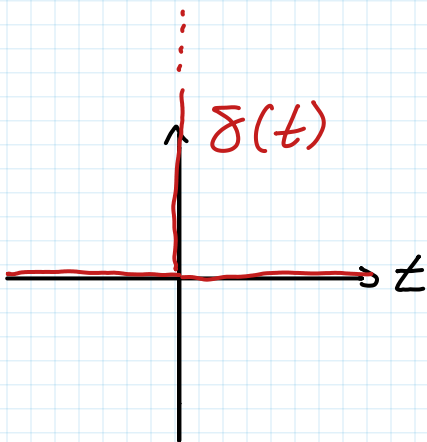
Position of poles will tell us about stability, asymptotic behaviour and oscillatory behavior!

To visualize the effect of the position of poles and zeros, we will use two fundamental input signals:  $\Rightarrow$  Test-signals

Test-Signals (some examples)

• Dirac impulse:  $\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & \text{else} \end{cases}$

• Heaviside function:  $h(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$   
(step function)



$\Rightarrow$  Why are they relevant? Let's look at  $\delta(t)$ :

• The reaction of a system to a  $\delta(t)$  input is called impulse-response:

$$Y(s) = G(s)U(s) = G(s) \cdot \cancel{\mathcal{L}\{\delta(t)\}}$$

$$\Rightarrow Y(s) = G(s) \quad | \quad \mathcal{L}^{-1}\{\cdot\}$$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\{G(s)\}$$

How do we find this?

| $x(t)$                      | $X(s)$                          |
|-----------------------------|---------------------------------|
| impulse: $\delta(t)$        | 1                               |
| step: $h(t)$                | $\frac{1}{s}$                   |
| $h(t) \cdot e^{at}$         | $\frac{1}{s-a}$                 |
| $h(t) \cdot \sin(\omega t)$ | $\frac{\omega}{s^2 + \omega^2}$ |
| $h(t) \cdot \cos(\omega t)$ | $\frac{s}{s^2 + \omega^2}$      |

• Partial fraction expansion

$$G(s) = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \dots + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_0}$$

$$G(s) = \frac{r_1}{s-p_1} + \frac{r_2}{s-p_2} + \dots + \frac{r_n}{s-p_n}$$

$r_i$ : residues

Of these terms we can easily find  $\mathcal{L}^{-1}\{\cdot\}$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\left\{\frac{r_1}{s-p_1} + \frac{r_2}{s-p_2} + \dots + \frac{r_n}{s-p_n}\right\} \quad | \quad \text{Laplace is linear!}$$

$$\Rightarrow y(t) = r_1 e^{p_1 t} + r_2 e^{p_2 t} + \dots + r_n e^{p_n t}$$

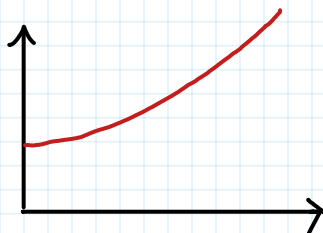
Now we can think of the effect of different poles.

Recall:  $s = \sigma + j\omega$

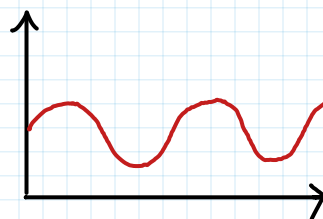
•  $e^{\sigma t}$ ,  $\sigma < 0$



•  $e^{\sigma t}$ ,  $\sigma > 0$



•  $e^{j\omega t} \pm e^{-j\omega t}$



$\Rightarrow$  Matlab

## Zeros

To understand the effect of zeros, consider this:

$$u(t) \rightarrow \boxed{\frac{d}{dt}} \rightarrow y(t)$$

$$y(t) = \frac{d}{dt} u(t)$$

$\Downarrow$

Recall this property:

$$\boxed{\mathcal{L}\{f^{(n)}(t)\} = s^n F(s)}$$

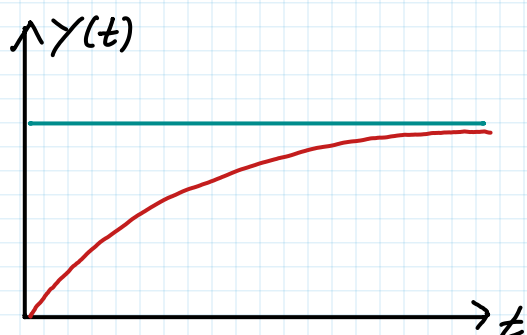
$$\left. \begin{array}{l} Y(s) = s \cdot \underbrace{U(s)} \\ \text{New zero in TF!} \end{array} \right\}$$

$\Rightarrow$  Zeros have some connection to derivatives...

How does adding zeros to a system affect their responses?

• Let's try and see this with step responses

- Assume some **system**  $G(s)$  without zeros and as input, apply a **step**  $h(t)$ :



=> How would this change if we added a zero?

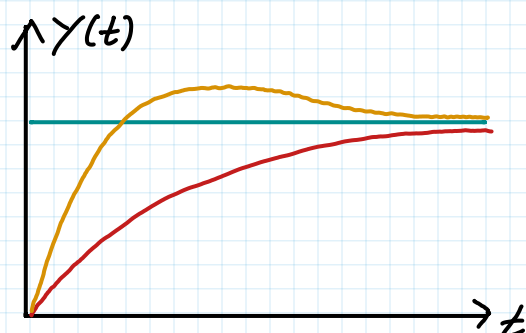
$$\tilde{G}(s) = \left( \frac{s}{-\frac{1}{2}} + 1 \right) G(s)$$

$$= -\frac{s}{\frac{1}{2}} G(s) + G(s)$$

=> The "s" acts as a derivative so the output would look like:

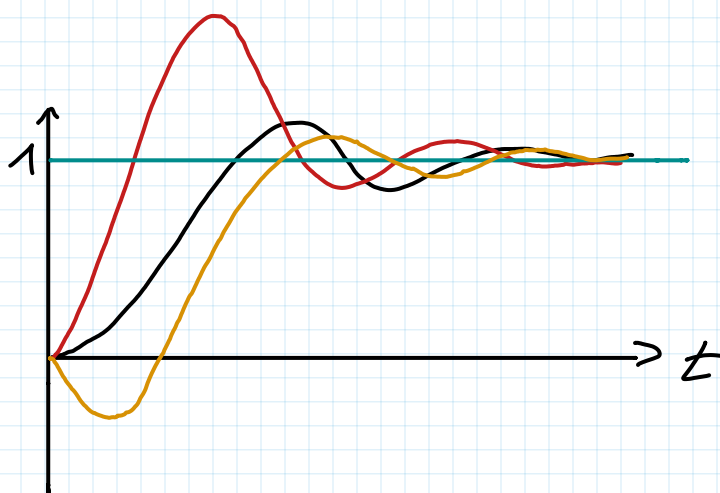
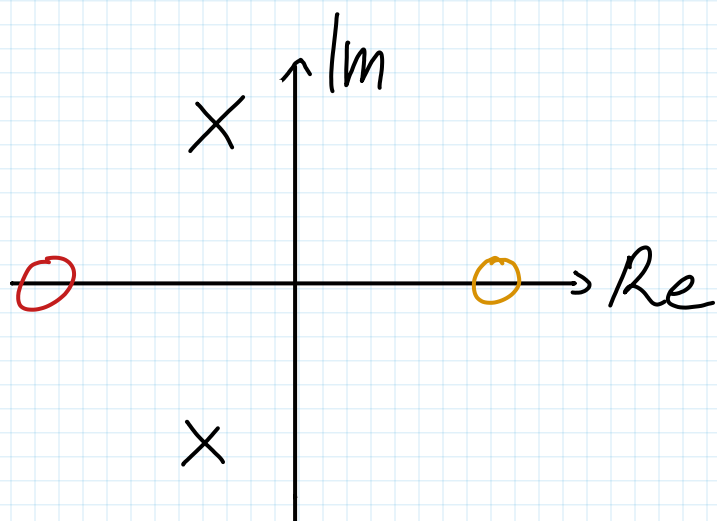
$$\tilde{y}(t) = -\frac{1}{2} \dot{y}(t) + y(t)$$

We add on top a scaled derivative of  $y(t)$ !



The zero has anticipatory effect! It "knows in advance" that it is off from the goal.

- Placement of zeros



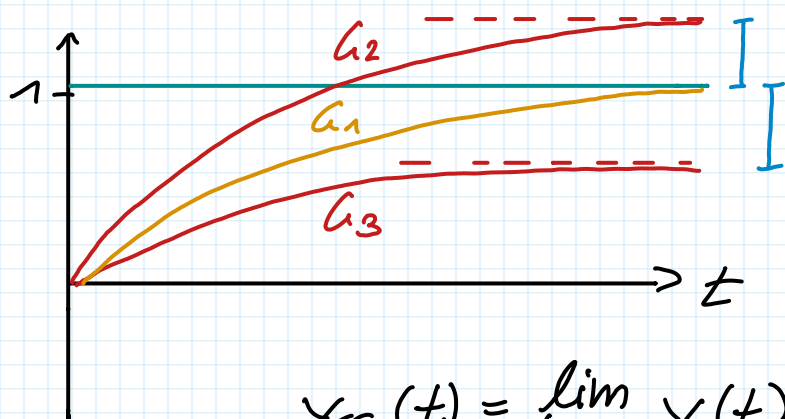
- Zeros in left-half plane: **Minimumphase zeros**
- Zeros in right-half plane: **Non-minimumphase zeros**

↳ Think of how the location of  $z$  affects

$$\tilde{y}(t) = -\frac{1}{2}\dot{y}(t) + y(t)$$

- The step response is a really useful tool to analyze systems.

⇒ One thing that we have glossed over...



- A system doesn't necessarily converge to the input value!

$$y_{ss}(t) = \lim_{t \rightarrow \infty} y(t) \neq u(t)$$

- So how could we quantify this **offset**?

## Final Value Theorem

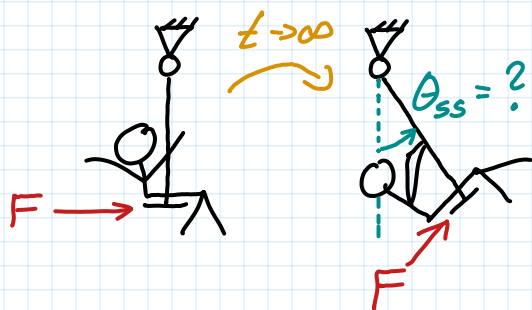
$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} s Y(s) = \lim_{s \rightarrow 0} s G(s) U(s)$$

- Step function:  $y_{ss} = \lim_{s \rightarrow 0} s \cdot G(s) \cdot \overbrace{H(s)}^{1/s}$

$$y_{ss} = \lim_{s \rightarrow 0} \cancel{s} \cdot G(s) \cdot \cancel{1/s} = \lim_{s \rightarrow 0} G(s)$$

| $x(t)$                      | $X(s)$                          |
|-----------------------------|---------------------------------|
| impulse: $\delta(t)$        | 1                               |
| step: $h(t)$                | $\frac{1}{s}$                   |
| $h(t) \cdot e^{at}$         | $\frac{1}{s-a}$                 |
| $h(t) \cdot \sin(\omega t)$ | $\frac{\omega}{s^2 + \omega^2}$ |
| $h(t) \cdot \cos(\omega t)$ | $\frac{s}{s^2 + \omega^2}$      |

Example: Step force applied to swing



$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ g/l & -\frac{c}{ml^2} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \frac{1}{ml} \end{bmatrix} u(t)$$

$$y(t) = [1 \ 0] x(t)$$

$$F(t) = \begin{cases} 0, & t < 0 \\ 1, & \text{else} \end{cases}$$

• We could now solve this as

$$\theta_{ss} = \lim_{t \rightarrow \infty} \left( C e^{At} x_0 + \int_0^t C e^{A(t-\tau)} B d\tau \right)$$

Good luck....

Better idea: transform to frequency domain!

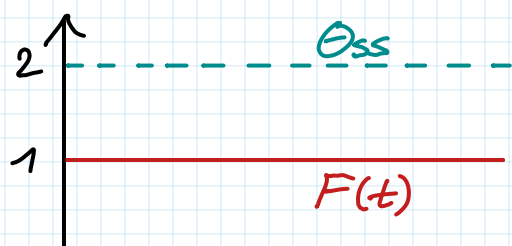
$$G(s) = C(sI - A)^{-1}B + D = \frac{4}{s^2 + s + 2}$$

Random params

Then use final value theorem

$$\theta_{ss} = \lim_{s \rightarrow 0} \cancel{s} G(s) \cancel{U(s)}, \quad U(s) = H(s) = \frac{1}{s} \quad (\text{step-function})$$

$$= \lim_{s \rightarrow 0} \frac{4}{s^2 + s + 2} = \frac{4}{2} = 2 \quad (\text{Degrees, radians, ...})$$



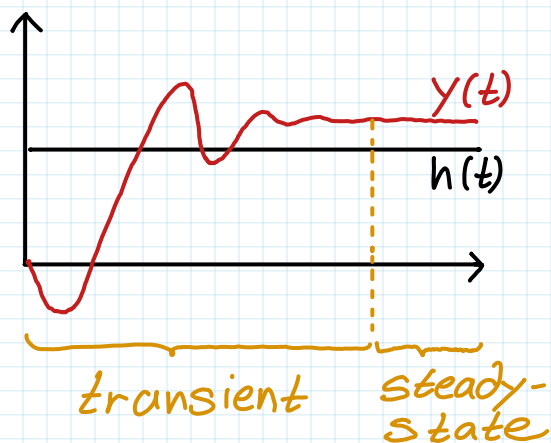
This factor has a name..

$$\Rightarrow G(s \rightarrow 0) = K_{\text{Bode}}$$

"Bode gain"

## In summary:

=> If you have a system  $G(s)$  and want to know how it responds to a step/impulse

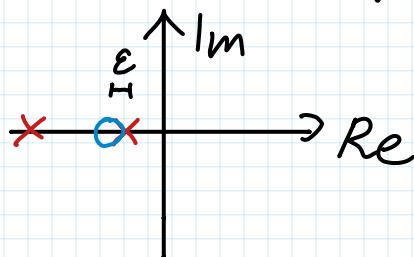


Transient: Poles/zeros  
Steady-state:  $k_{Bode}$

## Pole-zero cancellation

• Consider a system with a zero near a pole

$$G(s) = \frac{s+1+\varepsilon}{(s+1)(s+2)}$$



what happens if  $\varepsilon \rightarrow 0$ ?

$$G(s) = \frac{\cancel{s+1}}{(\cancel{s+1})(s+2)} \quad \left\{ \begin{array}{l} \text{The pole and the zero seem} \\ \text{to cancel!} \end{array} \right.$$

## But they don't!!!

The internal dynamics remain, we just don't see them anymore in the output! Dangerous if unstable.

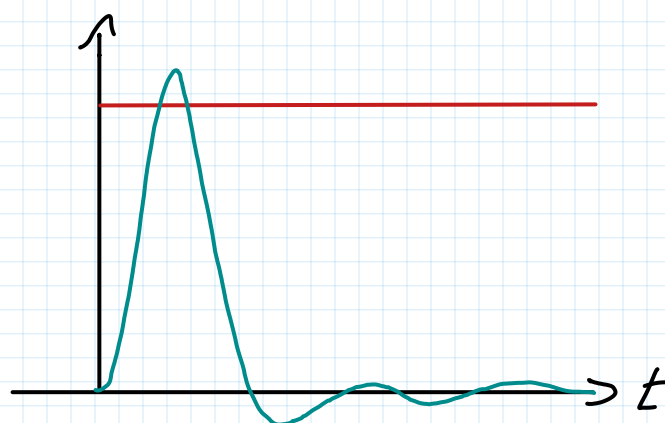
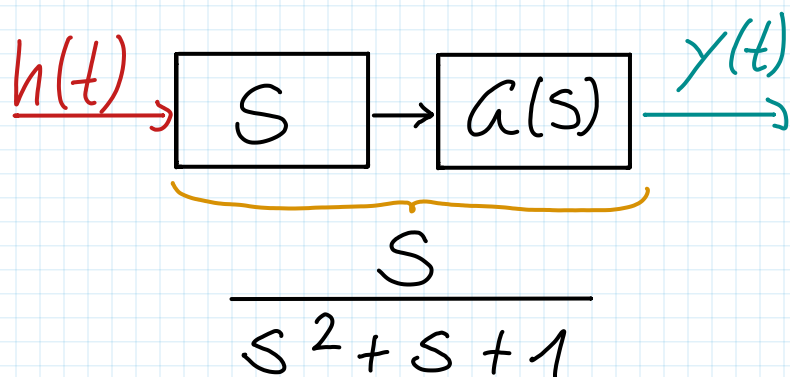
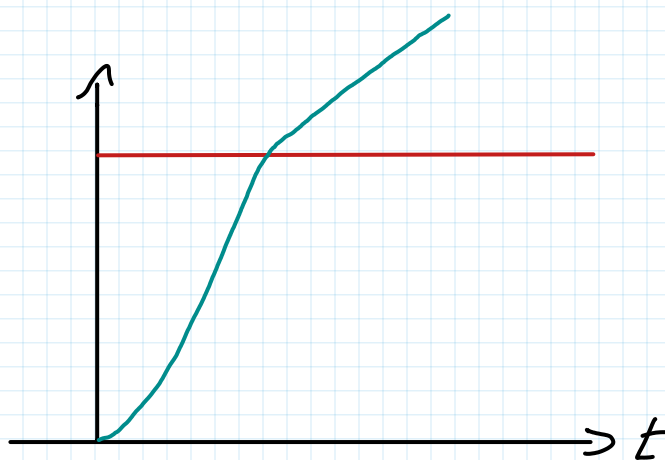
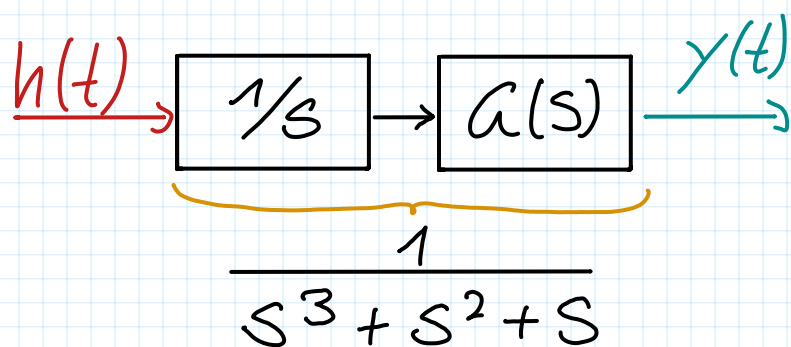
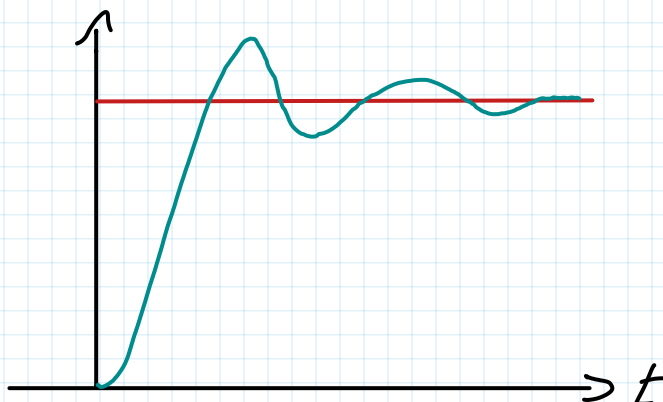
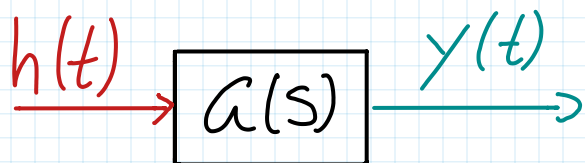
## Final note

Poles at  $p=0$ : Integrators

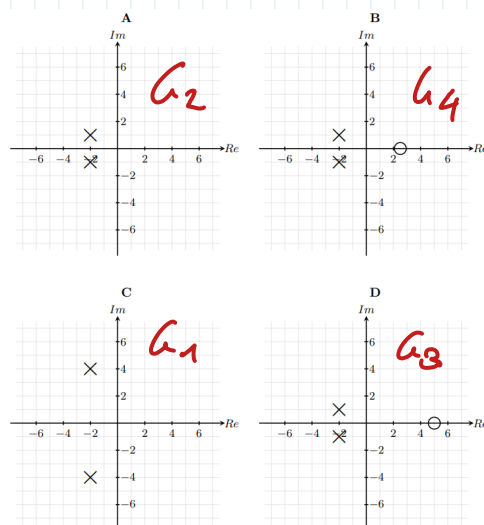
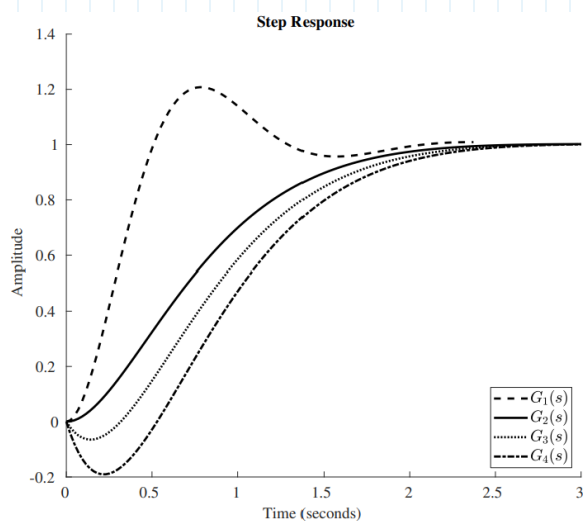
Zeros at  $z=0$ : Differentiators

$\Rightarrow$  We will use this in controller design

$$G(s) = \frac{1}{s^2 + s + 1}$$



## Q1) Match poles/zeros to step response



$$\tilde{y} = y + \frac{1}{-2} \dot{y}$$

## Q2)

**Problem:** Consider a linear time-invariant system with transfer function  $G(s)$  that is given by a root-locus gain  $k_{r1} = 1$  and by two poles and two zeros as shown in Figure 7.

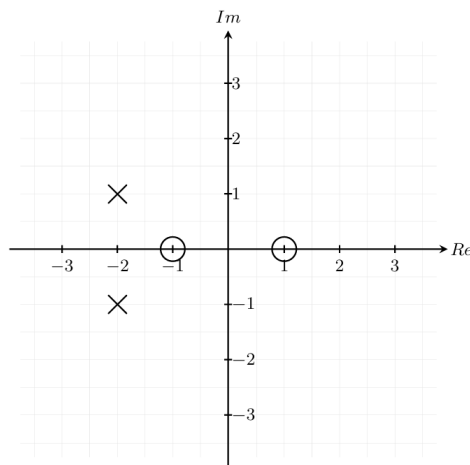


Figure 7: Poles ( $\times$ ) and zeros ( $\circ$ ) of the LTI system  $G(s)$ .

**Q19 (1 Points)** Mark the correct answer.

What is the magnitude  $|G(j)|$  of  $G(s = j)$ ?

- ☒  $|G(j)| = \frac{1}{2\sqrt{2}}$     ☐  $|G(j)| = \frac{1}{4}$     ☐  $|G(j)| = \frac{1}{2}$     ☐  $|G(j)| = \frac{1}{\sqrt{2}}$

**Q20 (1 Points)** Mark the correct answer.

What is the phase  $\angle G(j)$  of  $G(s = j)$ ?

- ☐  $\angle G(j) = 45^\circ$   
☐  $\angle G(j) = 0^\circ$   
☐  $\angle G(j) = -45^\circ$   
☒  $\angle G(j) = 135^\circ$

Solution:

Poles:  $-2+j, -2-j$

Zeros:  $1, -1$

$$G(s) = \frac{(s-1)(s+1)}{(s+2-j)(s+2+j)}$$

$$G(s=j) = \frac{(j-1)(j+1)}{2(2+2j)} = \frac{-2}{4+4j} = -\frac{1}{2+2j}$$

$$\cdot |G(s=j)| = \frac{|-1|}{|2+2j|} = \frac{1}{\sqrt{(2)^2+(2)^2}} = \frac{1}{\sqrt{8}} = \frac{1}{2\sqrt{2}}$$

$$\cdot \angle G(s=j) = \angle -\frac{1}{2+2j} = \angle -1 + \angle \frac{1}{2+2j}$$

$$= \pi + \cancel{\angle -1} - \angle 2+2j = \pi - \tan^{-1}\left(\frac{2}{2}\right)$$

$$= \pi - \frac{\pi}{4} = \frac{3\pi}{4} = 135^\circ$$